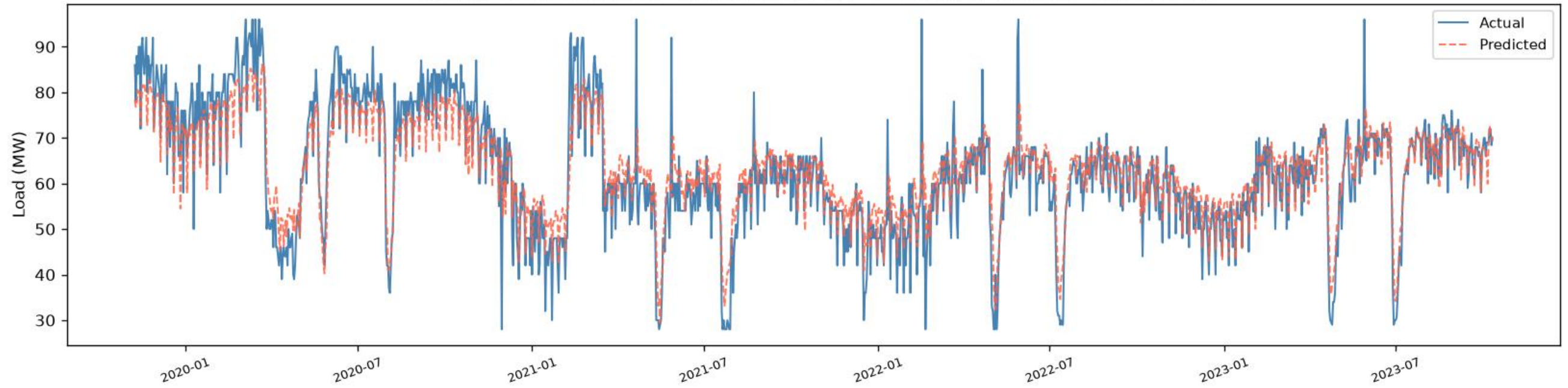
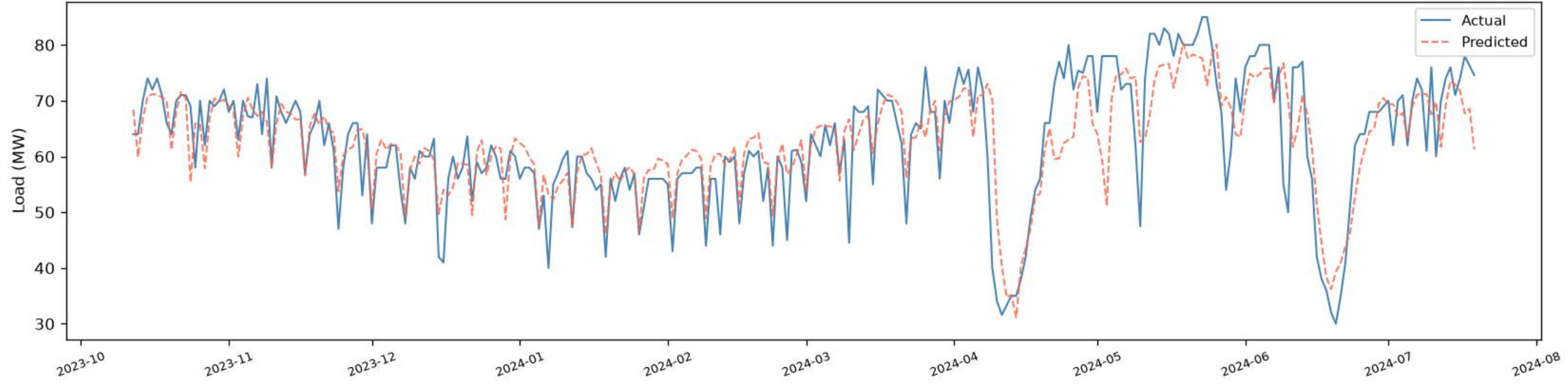


Matual - CNN+BiGRU+BiLSTM: Actual vs Predicted

Matual | CNN+BiGRU+BiLSTM - TRAIN (fit block) (R2=0.8021, RMSE=6.04, MAE=4.47, n=1398)



Matual | CNN+BiGRU+BiLSTM - VALIDATION (chronological, before test) (R2=0.6650, RMSE=6.52, MAE=4.63, n=272)



Matual | CNN+BiGRU+BiLSTM - TEST (≥ 2025), one-day-ahead (R2=0.8075, RMSE=5.70, MAE=4.24, n=499)

